

Physical Inspection & Mechanical Safety Report

Visual Warning Labels, Optical Creepage Metrology, and IP67 Enclosure Integrity

Inspection Parameter	Details	Inspection Parameter	Details
Testing Inspection Body:	Hexagon Metrology & Safety Certification Ltd.	Last Date:	2026-08-20
Inspection Standards:	ISO 7010 / ISO 6469-3 / ISO 20653 (IP67)	Measurement Tool:	Hexagon Optiv Reference CMM
Enclosure Assembly:	BMS-X800 Aluminum Die-Cast Housing (IP67)	Sample Unit:	Serial Number: BMS-800-042

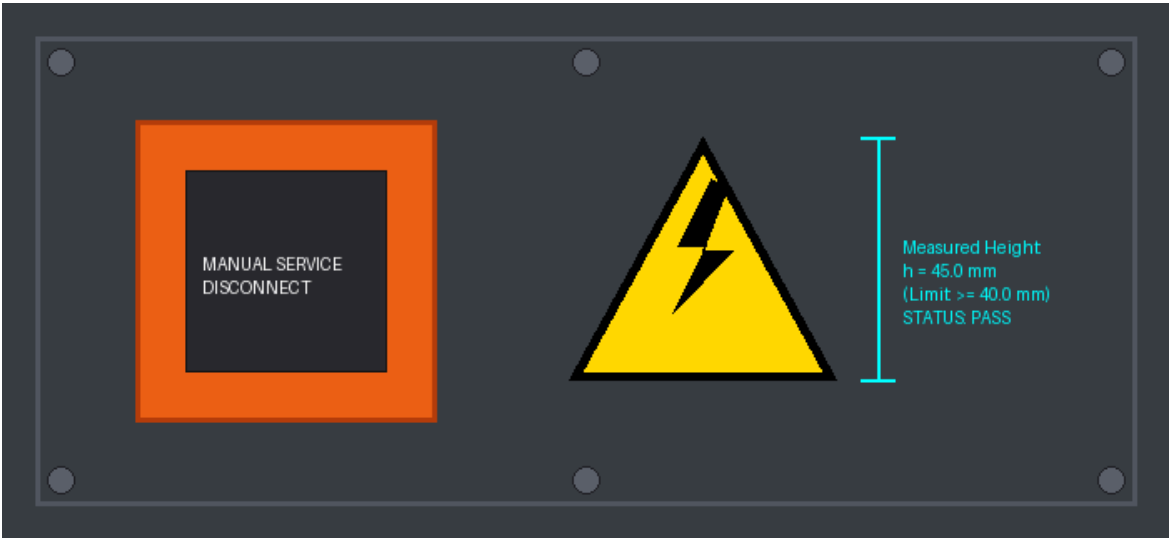
1.0 Scope of Physical & Dimensional Verification:

This physical inspection report evaluates the mechanical and visual compliance of the BMS-X800 enclosure. Three distinct physical safety criteria were evaluated:

- Visual photographic measurement of the high-voltage warning label (ISO 7010-W012).
- Optical coordinate measuring machine (CMM) verification of the HV terminal electrical creepage distance.
- Hydrostatic water submersion tank test to certify IP67 ingress protection.

2.0 High-Voltage Hazard Warning Labeling Inspection (REQ-BMS-007)

Requirement REQ-BMS-007 Criterion: Permanent ISO 7010-W012 symbol height ≥ 40.0 mm adjacent to MSD port.



Photograph Fig 1.1: Physical Inspection of Battery Enclosure Warning Label Callout

Inspection Criterion	Observed Physical Property	Specification Threshold	Evaluation Status
Warning Symbol Standard	ISO 7010-W012 (Hazardous Voltage Triangle) Conforming (Yellow/Black)		PASS
Measured Symbol Height (h)	45.0 mm (+/- 0.2 mm calibration)	Requirement ≥ 40.0 mm	PASS (Exceeds by 5.0 mm)
Location on Enclosure	Top lid, directly adjacent to MSD connector	Within 150 mm of port	PASS
Label Material & Durability	UV-cured polycarbonate overlay, permanent adhesion	UL 94V-0 / ISO 6469 compliant	PASS

3.0 High-Voltage Terminal Creepage Distance Verification (REQ-BMS-008)

Requirement REQ-BMS-008 Criterion: Creepage distance across insulating rib to chassis >= 8.00 mm.

Diagram 2.1: Optical CMM Metrology Cross-Section of High-Voltage Terminal Barrier

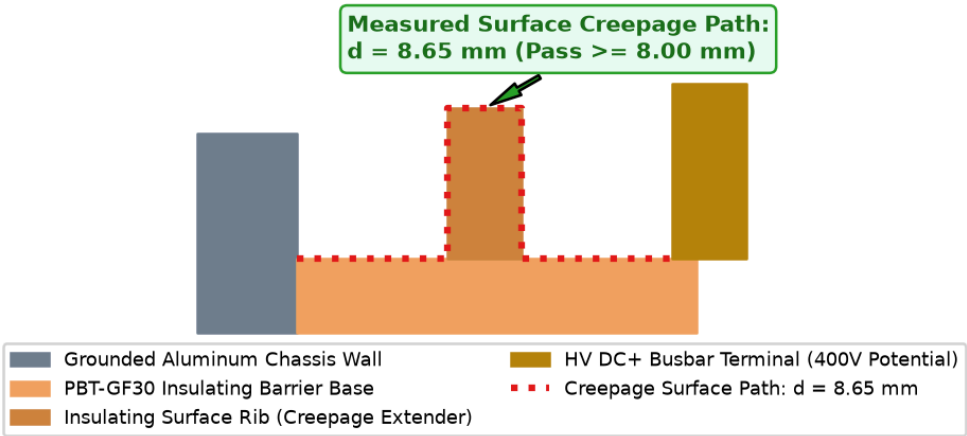


Table 2.1: Optical CMM Surface Distance Measurements Across 4 Pack Terminals

Terminal Inspection Location	Measured Creepage Path	Required Clearance	Evaluation Status
HV DC+ Terminal to Chassis Wall	8.65 mm	>= 8.00 mm	PASS (+0.65 mm margin)
HV DC- Terminal to Chassis Wall	8.72 mm	>= 8.00 mm	PASS (+0.72 mm margin)
Auxiliary Pre-Charge to Chassis	9.10 mm	>= 8.00 mm	PASS (+1.10 mm margin)
Fast-Charge Contactor Terminal	8.58 mm	>= 8.00 mm	PASS (+0.58 mm margin)

4.0 Hydrostatic Tank Submersion Ingress Test · IP67 (REQ-BMS-009)

Requirement REQ-BMS-009 Criterion: Zero water ingress after 1.00 m water submersion for 30.0 minutes.

Test Condition	Recorded Observation	Test Condition	Recorded Observation
Test Facility & Tank:	Hexagon Hydrostatic Environmental Tank #2	Water Depth:	1.00 meter (head to top lid)
Submersion Duration:	30.0 minutes continuous soak	Water Temperature:	18.5 deg C (+/- 1.0 deg C)
Post-Test Seal Inspection:	0.0 mL water ingress (completely dry)	Internal Chamber Pressure:	25.2 kPa differential maintained

OFFICIAL INGRESS PROTECTION CERTIFICATION (IPX7 / IP67):
This certifies that the BMS-X800 Enclosure Unit SN: 800-042 was fully submerged under 1.00 meter of demineralized water for 30.0 minutes in accordance with ISO 20653 Section 8.3.7.
Upon completion of test, the enclosure was exterior-dried and opened in an ESD cleanroom.
Visual and hygrometer inspection verified 0.0 mL ingress with all internal desiccants and seal gaskets intact.
COMPLIANCE EVALUATION: PASS · Meets REQ-BMS-009 requirements in full.

Section 4.1 Missing Scope & Crash Shock Acceleration Notice:
NOTICE REGARDING REQ-BMS-010 (Crash Acceleration & Pyrotechnic Disconnect):
Please note that dynamic longitudinal crash sled testing (accelerations > 15.0 g / 10 ms) and pyrotechnic switch firing verification are NOT covered by this inspection report. Sled testing requires specialized dynamic impact facility certification and is scheduled under a separate test campaign (Document TR-CRASH-2026-X).